

Sequential Telemetry and the Contraction of Certified Reachable Sets

Ethan Knox

Independent Researcher, Champaign, IL, USA

2026

Abstract

Keywords: reachable sets; occupation measures; set-membership estimation; data assimilation; Koopman operator; ultraspherical spectral methods; hybrid systems; atmospheric entry

MSC 2020: [select five from the list in the source]

Contents

1	Introduction	5
1.1	Static certified bounds and their blind spot	5
1.2	Set-membership estimation	5
1.3	Thesis statement	5
1.4	Outline of contributions	5
2	The Sequential Problem	6
2.1	Inherited formulation	6
2.2	Measurement epochs and the observation model	6
2.3	What a bounded-error return certifies	6
2.4	The sequential reachability problem	6
2.5	What contraction can and cannot mean	6
3	Measure Restriction and the Conditioned Problem	7
3.1	Restriction of an occupation measure	7
3.2	The conditioned Liouville problem	7
3.3	Piecewise propagation and re-initialization	7
3.4	The wrapping effect	7
3.5	Restriction on the reduced object	7
4	Contraction	8
4.1	Monotonicity under support restriction	8
4.2	One-shot contraction	8
4.3	Sequential contraction: the competition	8
4.4	Informativeness: when contraction is strict	8
4.5	Limits	8
5	Closed-Form Update of the Certificate	9
5.1	What restriction does to the certificate	9
5.2	The update map	9
5.3	Exact arithmetic across epochs	9
5.4	Cost per epoch	9
5.5	Numerical cross-check by spectral propagation	9
6	Verification and Results	10
6.1	Recovering the one-shot bound	10
6.2	Contraction under an informative return sequence	10
6.3	The competition: measurement against wrapping	10
6.4	Cost per epoch	10
6.5	Sensitivity to the error-bound assumption	10
6.6	Reproducibility environment	10
7	Discussion	11
7.1	Limitations	11
7.2	Relation to observer design	11
7.3	Scope	11
7.4	Extensions	11

A	Observation Models as Semialgebraic Sets	13
B	Banded Propagation Algebra	13
B.1	The generator in the ultraspherical basis	13
B.2	Truncation and its error	13
B.3	Cost model	13
C	The Wrapping Effect	13
C.1	Wrapping for polynomial superlevel sets	13
C.2	Per-epoch inflation bound	13
C.3	Mitigations	13
D	Proof of the Contraction Theorem	13
D.1	Monotonicity under restriction	13
D.2	The sequential composition	13

Open items

everything here is conditional on paper 1: the multi-phase occupation-measure embedding and the derived certificate (Φ, Ψ)	6
make-or-break condition for this paper: per "epoch", does measurement restriction contract faster than wrapping inflates? if wrapping wins there is no contraction theorem (bound the number of "epochs" over which contraction is strict?).	8

1 Introduction

1.1 Static certified bounds and their blind spot

1.2 Set-membership estimation

1.3 Thesis statement

1.4 Outline of contributions

2 The Sequential Problem

2.1 Inherited formulation

everything here is conditional on paper 1: the multi-phase occupation-measure embedding and the derived certificate (Φ, Ψ) .

2.2 Measurement epochs and the observation model

2.3 What a bounded-error return certifies

2.4 The sequential reachability problem

2.5 What contraction can and cannot mean

3 Measure Restriction and the Conditioned Problem

- 3.1 Restriction of an occupation measure
- 3.2 The conditioned Liouville problem
- 3.3 Piecewise propagation and re-initialization
- 3.4 The wrapping effect
- 3.5 Restriction on the reduced object

4 Contraction

4.1 Monotonicity under support restriction

4.2 One-shot contraction

4.3 Sequential contraction: the competition

make-or-break condition for this paper: per "epoch", does measurement restriction contract faster than wrapping inflates? if wrapping wins there is no contraction theorem (bound the number of "epochs" over which contraction is strict?).

4.4 Informativeness: when contraction is strict

4.5 Limits

5 Closed-Form Update of the Certificate

5.1 What restriction does to the certificate

5.2 The update map

5.3 Exact arithmetic across epochs

5.4 Cost per epoch

5.5 Numerical cross-check by spectral propagation

6 Verification and Results

- 6.1 Recovering the one-shot bound
- 6.2 Contraction under an informative return sequence
- 6.3 The competition: measurement against wrapping
- 6.4 Cost per epoch
- 6.5 Sensitivity to the error-bound assumption
- 6.6 Reproducibility environment

7 Discussion

7.1 Limitations

7.2 Relation to observer design

7.3 Scope

7.4 Extensions

References

A Observation Models as Semialgebraic Sets

B Banded Propagation Algebra

B.1 The generator in the ultraspherical basis

B.2 Truncation and its error

B.3 Cost model

C The Wrapping Effect

C.1 Wrapping for polynomial superlevel sets

C.2 Per-epoch inflation bound

C.3 Mitigations

D Proof of the Contraction Theorem

D.1 Monotonicity under restriction

D.2 The sequential composition